

Phishing / Network Security ML Pipeline

Model Training Results & Experiment Tracking Summary

Step 1 — Running the Training Pipeline

Executed `python main.py` from the terminal. The training pipeline runs end-to-end, trained models and artifacts are generated inside the **Artifacts** directory, and results are automatically pushed to **Dagshub**.

On Dagshub, navigating to **Experiments** shows the latest **7 logged runs** — 5 classifier comparison runs, plus 2 runs capturing the best model's performance on the training and testing datasets respectively.

```
=== Model Comparison ===
Random Forest: 0.9711
Decision Tree: 0.9649
Gradient Boosting: 0.9635
Logistic Regression: 0.9352
AdaBoost: 0.9320
```

Console output — model comparison scores (F1) across all 5 classifiers

BikashBIOS / Network-Security

Connected to https://github.com/BikashBIOS/Network-Security.git - 4 hours ago

Files

Datasets

Experiments 49

Models

Annotations

Collaboration

Settings

Unwatch 1

Star

Fork

Compare

Reset filters

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Labels

Columns

Go to MLflow UI

Log Experiment

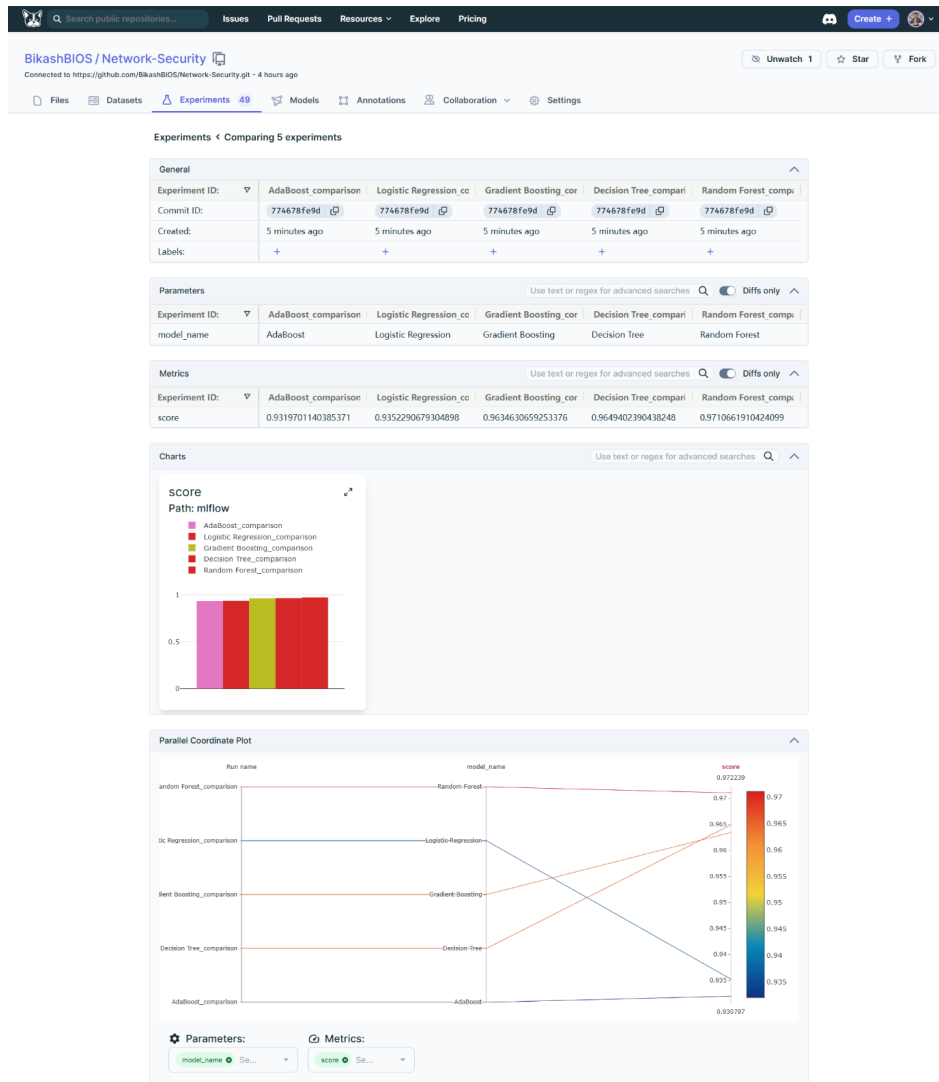
	Code	Data	Name	Commit	Created	Labels	model name	f1 score	precision	recall score	score
☐			skittish-hare-418	7746	2 minutes ago	+		0.971066191...	0.960784313...	0.981570512...	
☐			adventurous-yak-78	7746	3 minutes ago	+		0.992068334...	0.990456852...	0.993685068...	
☐			AdaBoost_comparison	7746	3 minutes ago	+	AdaBoost				0.931970114...
☐			Logistic Regression_comparison	7746	3 minutes ago	+	Logistic Regr...				0.935229067...
☐			Gradient Boosting_comparison	7746	3 minutes ago	+	Gradient Boo...				0.963463065...
☐			Decision Tree_comparison	7746	3 minutes ago	+	Decision Tree				0.964940239...
☐			Random Forest_comparison	7746	4 minutes ago	+	Random Forest				0.971066191...

Dagshub Experiments tab — 7 tracked runs from the latest pipeline execution

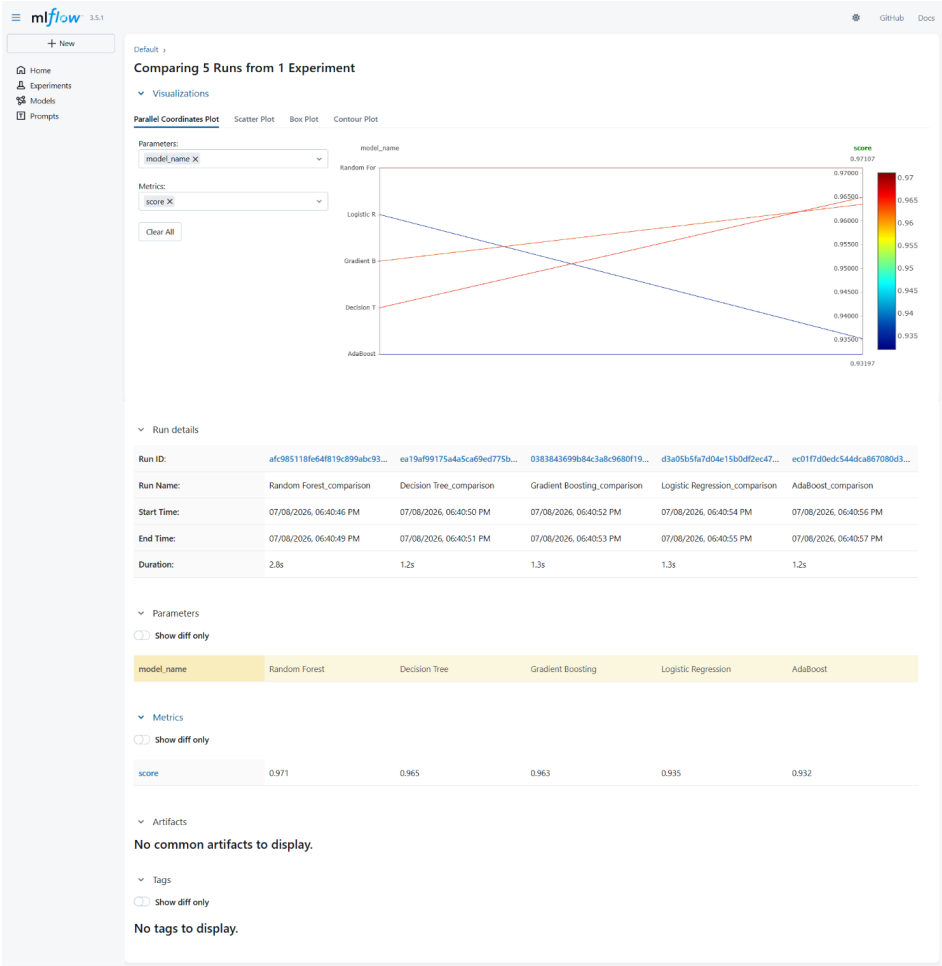
Step 2 — Verifying F1 Score, Precision & Recall

Model performance metrics were cross-verified across two independent tracking sources — **Dagshub** and **MLflow** — to confirm consistency of the logged experiment data.

Dagshub



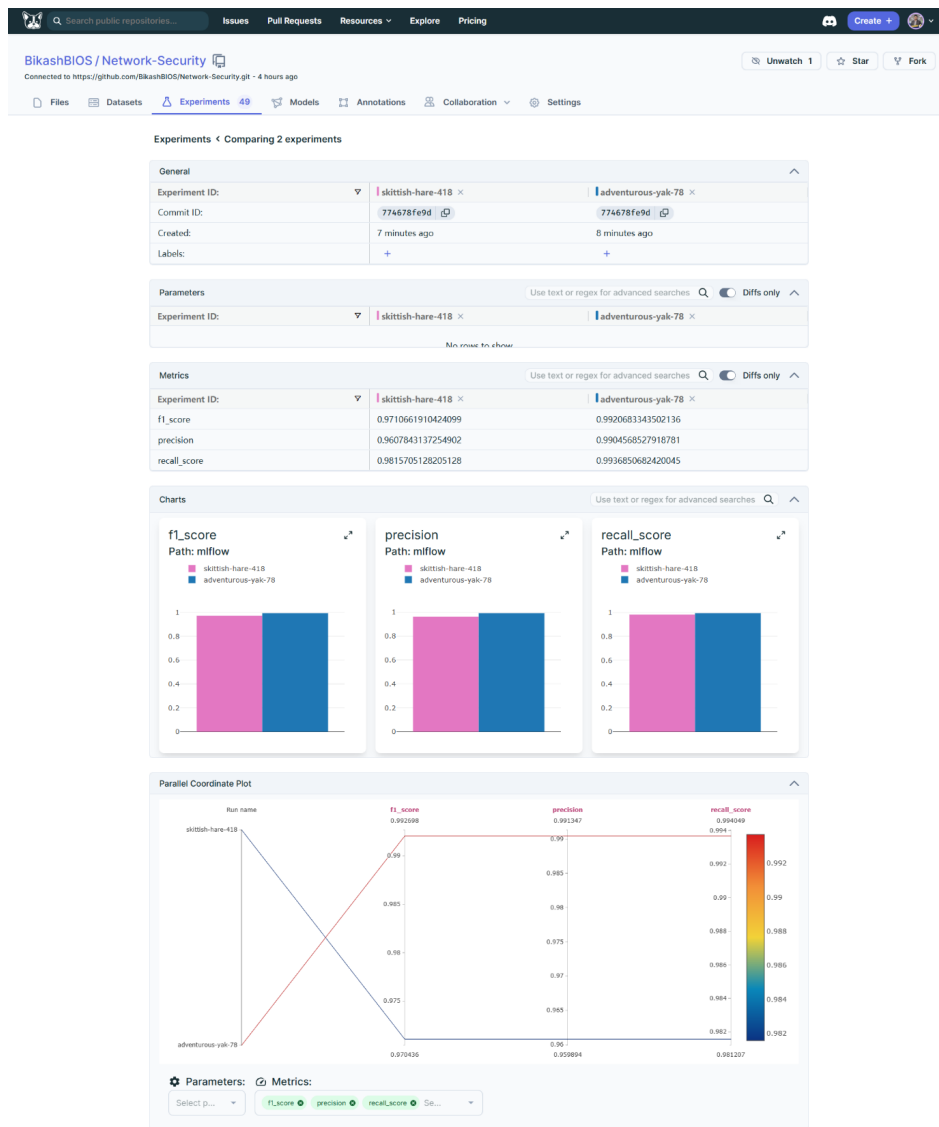
Dagshub — logged F1 score, precision, and recall across experiment runs



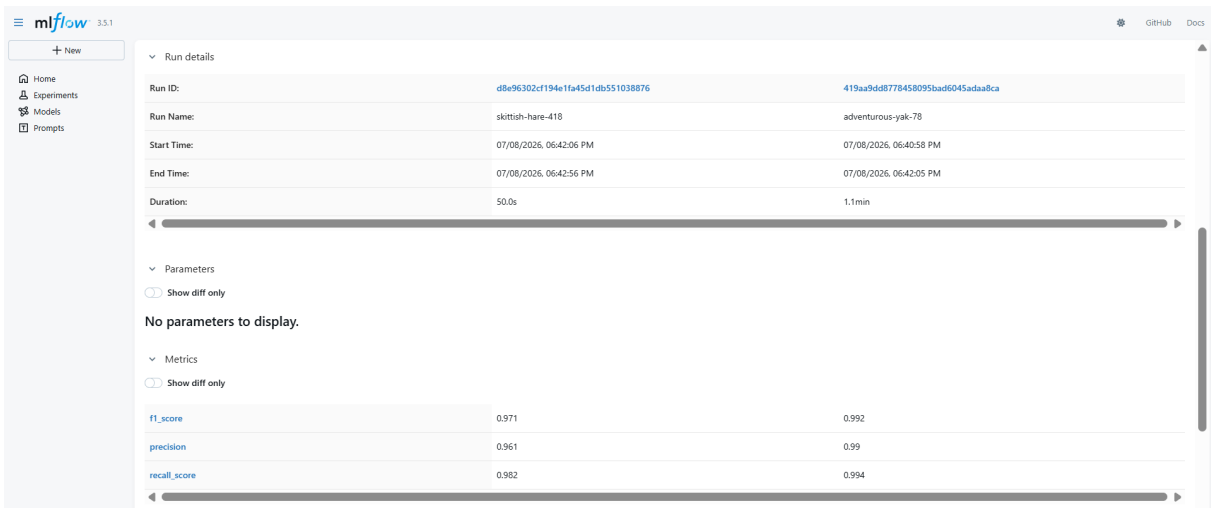
MLflow UI — matching metrics view, confirming consistent experiment tracking

Step 3 — Comparing Best Model: Training vs. Testing

The winning model (**Random Forest**) was evaluated on both the training and testing splits. Comparing these two runs verifies the model generalizes well rather than overfitting to the training data.



Best model — side-by-side comparison of training vs. testing performance



Metric comparison chart — training vs. testing run

Model Comparison Summary (F1 Score)

Model	F1 Score	Rank
Random Forest	0.9711	■ Best
Decision Tree	0.9649	2nd
Gradient Boosting	0.9635	3rd
Logistic Regression	0.9352	4th
AdaBoost	0.9320	5th

Captured Logs

```
[ 2026-07-08 18:40:46,119 ] 94 root - INFO - Model Report: {'Random Forest': 0.9710661910424099, 'Decision Tree': 0.9649402390438248, 'Gradient Boosting': 0.9634630659253376, 'Logistic Regression': 0.9352290679304898, 'AdaBoost': 0.9319701140385371}

[ 2026-07-08 18:40:46,119 ] 103 root - INFO - Best Model: Random Forest with score 0.9710661910424099
```